BizOS Architecture

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A walkthrough for PMs: how a patient message becomes a safe, owner-approved reply.

BizOS in one minute

AI writes the reply, code enforces the limits, and the owner approves every send. For Harbor Smile Dental, our fictional 10-person demo practice, that means:

1. A patient writes in through the web form ("Can I get a refund for my $500 cleaning?").
2. Claude reads it, picks the right department (front desk or billing) and drafts a reply from that department's rules.
3. A piece of plain code, the **Policy Gate**, checks the draft against hard limits from the same rules files. A $500 refund is over the $200 cap, so it is blocked no matter what the AI said.
4. Everything that passes waits in the owner's **approval queue**. Nothing is sent until a person clicks Approve.

The owner runs the business by editing plain Markdown rules files. A change takes effect on the next message, with no restart and no developer.

A message's journey

Every reply passes the gate and the owner before it is sent



message journey · 8 steps, rules on the side

The AI only writes; the rules files feed both the drafter and the gate, and a blocked draft never reaches the owner's queue.

The five parts

Each part has one job, and only the last one can send anything.

Part	What it does	Who changes it	Where it lives
Rules files	One Markdown file per department (company, front desk, billing). Plain sentences tell the AI what to say; a `policy:` block at the top holds the hard limits, like "refunds over $200 are blocked".	The owner, in the dashboard's rules editor	`business/*.rules.md`, seeded from `templates/dental-office`
AI agents	A fast model (Haiku) picks the department. A stronger model (Sonnet) drafts the reply, quoting the rule each statement comes from and listing any missing info.	Siddharth (prompts)	`server/src/agents/`
Policy Gate	Plain code, no AI. Checks every draft: do the cited rules exist, do the quotes match, is any dollar amount over a limit? Verdict is pass, review or block.	Nobody at runtime: limits come only from the rules files	`server/src/gate.ts`
Approval queue	Every draft that isn't blocked waits here for the owner. Urgent escalations (an emergency, a lawyer) sit at the top. Approve or reject.	The owner, in the dashboard	`GET/POST /api/approvals`

Part	What it does	Who changes it	Where it lives
Outbox and audit log	Sends approved replies once, re-checking the gate at the moment of sending. The audit log records every message, draft, verdict and decision.	Nobody: append-only	SQLite tables `outbox` and `audit`

Everything runs as one Node server with one SQLite file, on the business's own machine. The owner's dashboard is served by the same server at `/`.

Safety guarantees

The one-line version for judges: **the model proposes, the gate disposes, the owner decides.** Each line below is backed by automated tests.

- **No auto-send.** Every reply needs the owner's Approve click.
- **Limits live in one place.** The gate reads its numbers only from the `policy:` blocks. Change the file and both the AI's instructions and the gate's check change together.
- **Fails closed.** A missing or broken rules file, an unreadable amount, or an action type the rules don't name all send the draft to review instead of letting it through.
- **No invented rules.** A draft that cites a rule that doesn't exist, or quotes words not in that rule, is held for review.
- **Approvals can't go stale.** An approval is tied to the exact draft and rules version the owner saw. If either changed, it is refused and the owner looks again.
- **The AI can't pick the recipient.** Replies go to the address the message came from, never one the model writes.
- **The AI can't send.** Agent code is not allowed to touch the outbox or approvals; a repo check (`claims.ini`) fails the build if it tries.
- **Sent once.** Each reply has a unique send key, so a retry or restart never sends it twice.
- **No secrets in the browser or repo.** The API key stays in a local `.env` file on the server.

Live today vs roadmap

The full message-to-reply loop works end to end with live Claude agents; 9 of the 10 DAY_PLAN acceptance checks pass on a fresh install.

Area	Live today	Roadmap
Channels	Web form in, web form reply out	Email, SMS and other channels (one adapter each)
Agents	Department router + reply drafter, re-run after a restart	Prompt tuning on the drafter (Siddharth, issue #29)
Rules	Dental template with company, front desk and billing files; live editing with version history	More business templates
Approval queue	Held drafts plus urgent escalations, server side	Escalation cards in the dashboard (Kevin, in progress)
Money actions	Refund, discount, fee waiver and payment-plan limits	Catching one refund split into several small ones (issue #7)
Cases	One case per incoming message	Threading follow-up messages into one case (issue #25)

The one acceptance check still open, escalations visible in the queue, is done on the server and waits on Kevin's dashboard change.

Glossary

Term	Meaning
Case	One customer conversation, created when a message arrives
Draft	A reply the AI (or a person) proposes for a case; each new one gets a version number
Rule ID	The label at the start of a rule line, like `REFUND-CAP`; drafts cite these

Term	Meaning
`policy:` block	The hard limits at the top of a rules file, read by the gate
Verdict	The gate's answer: **pass** (owner approves), **review** (owner looks closer), **block** (cannot be sent)
Held	Waiting in the approval queue
Escalated	Needs a person now (emergency, legal, identity); shown as urgent in the queue
Blocked	Over a hard limit; never in the queue, never sendable
Outbox	The send list; each approved reply is sent exactly once
Audit log	The permanent record of everything that happened to a case